# MG-GLAM plan for the DEGRACE companions

**Schedule, lens planes and extra on-the-fly outputs: one merged plan for review**

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`docs/degrace_companion_implementation_plan.md`

# What the plan covers and where it stands

## Two sources, one file

- Slack agreement with Baojiu (16 Oct): 42-output schedule, no particle snapshots, 12 lens planes per output.
- Extra-outputs brief (17 Oct): initial-field modes, late-time low-$k$ density and momentum modes, 1 % hashed subsample, $f_R$ at haloes.
- Merged into `docs/degrace_companion_implementation_plan.md`: ground rules, conventions, items I1–I11, tests, budget, phasing, Phase-0 questions, session notes.

**Done:** I1 (`cz ae4fa9d`), ECOSMOG metadata, 42-epoch cross-check. **Open:** I2–I11.

| item | what | track | status |
|---|---|---|---|
| I1 | exact-$z$ targets closer than 0.5 % | A | **done** |
| I2 | lens planes, 12 per output | A | open |
| I3 | schedule input + generator | A | open |
| I4 | step $\rightarrow$ epoch index file | A | open |
| I5 | per-output class list | A | open |
| I6 | configuration, provenance | A | open |
| I7 | A: $\delta_L(k)$ dump | B | route to decide |
| I8 | B: low-$k$ $\delta_m$, momentum modes | B | open |
| I9 | C: 1 % quantised subsample | B | open |
| I10 | D: $f_R$ at halo centres | B | open |
| I11 | E: tidal ladder | B | fallback only |

Track A: campaign readiness (cheap, needed for any run). Track B: extra outputs for single-index halo statistics.

Source: plan sections 0, 5, 10, 14

# Ground rules for every session

## Physics and inputs

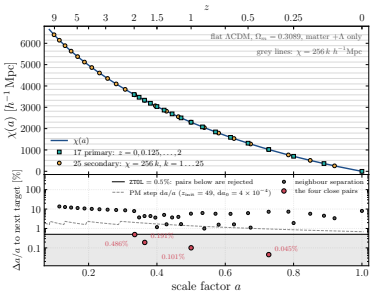
1. ICs are **always 2LPTIC**: the FML generator in `2LPTIC_Gui/`, or Baojiu's 2LPTic `ics.*` for matched-phase companions, through `ic2pm` (default `half`). $z_{\rm init} = 49$. `PMP2start` only to reproduce legacy runs.
2. Stepping: $\mathrm{d}a_0 = 6 \times 10^{-4}$ for $\Delta x \leq 0.25\,h^{-1}\mathrm{Mpc}$ (163 steps, the legacy late-time ladder); $8 \times 10^{-4}$ acceptable at $\Delta x = 0.5$; $4 \times 10^{-4}$ for a converged reference.
3. No change to the PM solver, multigrid, stepping, BDM or existing output formats.

## Engineering

4. Every addition behind a runtime flag, default off; with all flags off the code is bit-identical to `cz`.
5. Branch from `cz` (the mainline, not `main`); new code in new files, minimal call sites.
6. No new $N_g^3$ allocations unless the memory budget allows (f(R) at $N_g = 4096$ needs 950+ GB). Reuse `FI` at the output step.
7. Tests through Slurm on compute nodes; ask before jobs above ∼50 node-hours or production size.
8. Test boxes: $36^3$ (`test/precise_z`) for schedule logic; $256^3$, $L = 256$, $N_g = 512$, ΛCDM + F5 same seed, for the products.
9. After each phase, append what was done and the test numbers to the plan's section 16.

Source: plan section 0; `AGENTS.md`; `2LPTIC_Gui/VALIDATION.md`

# Output design and the numbers already verified




17 primary $z$ + 25 stations $\chi = 256\,k\,h^{-1}\mathrm{Mpc}$; four pairs closer than 0.5 % in $a$.

## Agreed with Baojiu

No particle snapshots (200 GB each at $2048^3$); velocities in the BDM catalogues and the subsample. Lens planes: 3 axes $\times$ 4 slabs of $256\,h^{-1}\mathrm{Mpc}$, $4096^2$ float32, 0.8 GB per output. 42 outputs; stations recomputed per cosmology.

## Verified

Baojiu's 42 scale factors = flat $\Lambda$CDM, $\Omega_m = 0.3089$, matter $+\Lambda$ only, to $8 \times 10^{-8}$; the same rule generates all 96 ECOSMOG lists ($2 \times 10^{-7}$). I1: close pairs accepted; a $3.3 \times 10^{-4}$ step costs one PM step and $\sim 2 \times 10^{-6}$ in velocity.

Companion scope: L1024 $N_g$ 4096 fiducial boxes ($2048^3$) and the 96 ECOSMOG nodes (Baojiu's $1024^3$ ICs).

Source: plan sections 1–2; `docs/degrace_ecosmog/`

# Shared conventions for every new product

## Fourier modes (I7, I8)

$\tilde{\delta}(\mathrm{k}) = N^{-3} \sum_x \delta(x)\, e^{-i\mathrm{k}\cdot\mathrm{x}}$, $\mathrm{k} = (2\pi/L)(i_x, i_y, i_z)$, $P = L^3 \langle |\tilde{\delta}|^2 \rangle$. One copy routine converts from FFTPACK5's layout for every dump (they are cross-correlated mode by mode). Store the cube $|i_x|, |i_y| \le n$, $0 \le i_z \le n$, $n = \lfloor k_c L/2\pi \rfloor$; $k = 0$ set to zero. Fortran `C(0:n,-n:n,-n:n)`; numpy `reshape(2n+1,2n+1,n+1)` → `[ix+n, iy+n, iz]`. $i_x$ is the catalogue's $x$ axis; `axis_map` recorded. No deconvolution inside GLAM.

## Velocity epoch

GLAM outputs are not synchronous: $x$ at $a$, $v$ at $a - \mathrm{d}a/2$ with the step just taken (down to $3.3 \times 10^{-4}$ in exact-$z$ mode). Every product with velocities uses the BDM pair and records `a_x`, `a_v`.

## Files

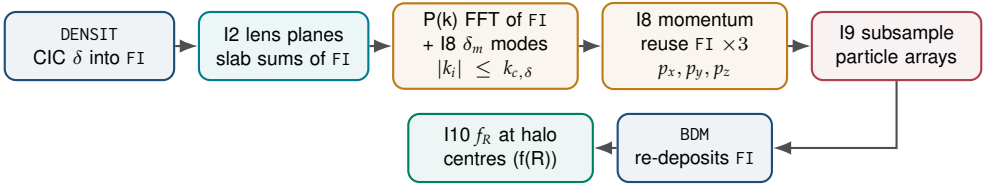
Raw little-endian binary, `access='stream'`, explicit `convert='little_endian'` (the build is big-endian), no record markers. Sidecar `.meta` with `key = value`: `git_hash`, `model`, `fR0`, `L`, `N_p`, `N_g`, `N_dep`, `a`, `z`, `a_x`, `a_v`, `step`, `box`, `units`, `axis_map`, `D_code(a)`, plus item keys. Names: `extra/<run>_a<a>_<product>.{bin,meta}` or GLAM's `<product>.<step>.<box>.dat`: decide once.

## Parameters

New flags appended to the end of `Init.dat` and mirrored in `Setup.dat` (`PMP2init`), read by `ReadSetup` with `iostat` and defaults off, so every existing run directory stays valid.

Source: plan section 3

# Sequence at an output step: one deposit serves everything




Today `Analysis` runs PDF → P(k) (`GetPower` FFTs `FI` in place, `mDENSIT=1`) → dump/project → bias → `BDM(mDENSIT)`. The new products slot in without a second deposit: lens planes and the $\delta_m$ modes use the same CIC field; the momentum deposits reuse `FI`, which nothing reads again before `BDM` (Phase 0, Q5, to confirm).

**Memory:** zero extra $N_g^3$ arrays with $N_{\rm dep} = N_g$; 0.8 GB for the planes; the mode cubes are $\leq 0.5$ GB. A separate $N_g/2$ buffer (34 GB) only if `sacct` MaxRSS shows headroom (Q8). `MG_scratch_release` already frees multigrid scratch around outputs.

Source: plan section 4; `PMP2mod_analyze.f90` `Analysis`, `PMP2mod_density.f90`, `PMP2mod_power.f90`

# Track A: campaign readiness (I2–I6)

## I2 Lens planes

Generalise `ProjectDENSIT`: per axis, mean $(1+\delta)$ of each slab column, optional block average, OpenMP as today. Flags `LENS_SLABS` (4), `LENS_GRID` (4096). One `.bin` per output, 12 planes $X_{1..4}$ $Y_{1..4}$ $Z_{1..4}$ + `.meta`. Logged: plane means $= 1$, slabs sum to the full projection.

## I3 Schedule input and generator

`#outputs = -42` + redshifts (never $z_{\rm init}$). Generator: ECOSMOG nodes from `aout.json/nodes.csv`; new cosmologies from $\chi(a)$ + the 17 primary $z$; writes the `Init.dat` block and the `PkTable.dat` header. Reproduces the fiducial list to $10^{-7}$, all 96 lists to $10^{-6}$.

## I4 Index file

`OutputSchedule.dat` at run start: `ISTEP a z chi exact class` per marked step; halocat maps snapshot numbers from it.

## I5 One per-output class list

`iBDM`, `iPower`, lens planes and `EXTRA_OUTPUT` must not all fire at 42 outputs. One class list (e.g. 1 = primary: BDM + P(k) + planes + extras; 2 = station: planes + P(k)) replaces the brief's `EXTRA_OUTPUT_LIST`. Default to agree with Baojiu.

## I6 Configuration and provenance

`iSave=0`, `iPower=1`, `iBDM=1`, planes on, `Pk tune = 1.0`, $z_{\rm init}$ 49, $\mathrm{d}a_0$ by the rule, `Ncheckpoint` so a lost 72 h job does not restart from $z_{\rm init}$. Record `Init.dat`, the index file, the `ic2pm` epoch line, `.meta` sidecars.

Source: plan section 5, I2–I6

# Track B: extra on-the-fly outputs (I7–I11)

## Why

Mass-pair $\xi_{hh}$ is shot-noise limited above $\log M \sim 14.5$: use halo $\times$ matter and halo $\times$ momentum cross-correlations instead. Store the matter side; keep mark, bin and rank definitions offline.

## I7 $\delta_L(k)$ (Item A)

Modes $|k_i| \leq 1.5$ of the initial Gaussian field, linear-theory normalised at $a = 1$ with the code's growth; identical files for f(R) and $\Lambda$CDM with the same seed. Gives a cosmic-variance-free halo propagator $\Gamma_h(k|M)$. Route to decide (next slide).

## I8 low-$k$ $\delta_m$ and $(1+\delta)$v (Item B)

$k_{c,\delta} = 1.5$ (3.0 if $R \approx 1\,h^{-1}\mathrm{Mpc}$ tidal fields are needed), $k_{c,p} = 1.0$; CIC on $N_{\rm dep} = N_g$ reusing `FI`. Files `deltam`, `mom_x,y,z`.

## I9 1 % quantised subsample (Item C)

Keep particle if `splitmix64(ID xor SALT) mod 100 == 0` (C via `iso_c_binding`, Python twin); increasing ID order, IDs implicit. int16 positions (15.6 kpc/h) and velocities (0.25 km/s): 12 B per particle, 1.03 GB at $2048^3$. Baojiu's `ics.*` are in lattice order with sequential IDs.

## I10 $f_R$ at halo centres (Item D)

Not recoverable offline: `fR_centre` (CIC at the BDM centre), `fR_mean` (within the BDM radius); float32, catalogue order, after BDM while the field is in memory.

**I11 (fallback):** tidal ladder $T_{ij}$, $R \in \{1, 2, 4, 8\}$, only if the I8 dump is too large. **Not in GLAM:** marks, binning, eigen-decomposition, Python readers.

Source: plan sections 5 (I7–I11) and 6

# Decision: the source of $\delta_L$ under the 2LPTIC rule

**Conflict**

The brief places the $\delta_L$ dump in `PMP2start`, but new runs never use `PMP2start`: the Gaussian field is generated by the FML generator (our boxes) or by Baojiu's 2LPTic (ECOSMOG companions).

**Route 1: FML generator**

Add the dump to `2LPTIC_Gui/` (our code): same copy convention and file format. Covers every new box. Cleanest; needs the generator's RNG and ordering recorded (Phase 0, Q1–Q2).

**Route 2: Baojiu's `ics.*`**

(a) derive $\delta_L = -\nabla\cdot\Psi$ from the IC displacements at ingest in `ic2pm`; the 2LPT term contaminates at the per-cent level near $k \sim 1$, negligible at $k < 0.1$ (quantify with V3); or (b) patch his 2LPTic for a regenerate-only dump with his `2LPTic.param` and seed 2026 (code and parameters on cosma7).

**Route 3: legacy pilots**

The brief's `IC_DUMP_ONLY` mode in `PMP2start` for the existing ZA-at-$z{=}100$ pilots: generate the field with the pilot's parameters and seed, write, exit. Needs the RNG thread-dependence answer.

Proposal: route 1 for new boxes and route 2(a) for the ECOSMOG companions, validated against 2(b) on one node; route 3 only if $\Gamma_h$ for the legacy pilots is still wanted. All three write the same `deltaL.{bin,meta}` and pass V1, V7, V8.

Source: plan section 5.17

# Validation and storage

| test | pass criterion |
|---|---|
| S1–S2 schedule | 65/65; generator to $10^{-7}$ / $10^{-6}$ |
| L1–L3 planes | means $= 1$; equals the offline projection; plane $P(k)$ vs 3-D $P(k)$ |
| V1 layout | $\cos$ / $\sin$ modes at $\pm m$, amplitude 1/2 |
| V2 $P(k)$ | `deltam` vs code $P_{mm}$ within $10^{-3}$ |
| V3 closure | `deltam`$(a_{\rm ini})$/`deltaL` $= D_{\rm code}$ to 0.5 %, $r > 0.999$ |
| V4 halo axes | $r \gtrsim 0.9$ at low $k$, no linear-in-$k$ phase |
| V5 subsample | $n_{\rm sel}$ within $3\sigma$; IDs, positions match; hashes agree |
| V6 momentum | subsample vs `mom_*` within shot noise |
| V7–V8 determinism | identical md5 (threads; f(R) vs $\Lambda$CDM) |
| V9 overhead | $< 5\,\%$ wall time; RSS within node memory |

| per output | $2048^3$ | $1024^3$ |
|---|---|---|
| lens planes $12 \times 4096^2$ | 0.8 GB | 0.8 GB |
| `CatshortV` | 0.3–0.4 GB | $\sim$0.1 GB |
| `deltam` ($k_c$ 1.5 / 3.0) | 0.47 / 3.7 GB | same |
| `mom_x,y,z` ($k_c$ 1.0) | 0.41 GB | same |
| subsample $M = 100$ | 1.03 GB | 0.13 GB |
| `deltaL` (once per run) | 0.47 GB | same |

**Budget decision**

Per $2048^3$ box: planes 34 GB, catalogues $\sim$15 GB, extras at all 42 outputs $\sim$80 GB ($\approx$130 GB per box, 13 TB for 100 boxes); extras at the 17 primary outputs only $\sim$32 GB ($\approx$80 GB, 8 TB). Baojiu's 3.4 TB covered lens planes alone.

Source: plan sections 8–9

# Phasing and open questions

**Order of work** (each phase: code, numbers, section-16 entry)

1. Phase 0 reconnaissance (plan section 13), no edits; report.
2. I3 generator + S2; I4 index; I5 class list.
3. I2 lens planes + L1–L3.
4. Pilot ECOSMOG node 2: ingest, run $\Lambda$CDM + f(R), compare $P(k)$ with the twins; runtime, `sacct`, sizes.
5. I7 (route decided) + V1, V7, V8; $\delta_L$ for one legacy pilot + V4.
6. I8 + V2–V4; 7. I9 + V5–V6; 8. I10 on the f(R) test box.
9. V9 production-size overhead run, after approval.

**For Baojiu**

BDM at all 42 or the 17 primary? Lens planes at primary epochs too? Plane normalisation (mean $1 + \delta$ vs surface density)? Keep both members of close pairs? Plus the ECOSMOG questions (CAMB $\omega_c$, r2g conversions, Node_900, $\Lambda$CDM $P(k)$).

**For CZ**

Extras cadence (all 42 vs primary only); $k_{c,\delta}$ 1.5 or 3.0; the $\delta_L$ route; file naming scheme; companion resolution ($N_g$ 2048 or 4096 on $1024^3$ particles); confirm the track order above.

Phase-0 facts already recorded in section 16: lattice order of Baojiu's ICs; $v$ at $a - \mathrm{d}a/2$ at outputs; `FI` life cycle in `Analysis`; `FI2`/`FI3` for MG; 950+ GB for f(R) at $N_g$ 4096; sequential `Init.dat` parser.

Source: plan sections 10, 11, 13, 16